

YOAN KESRAOUI

SALES AND TRADING | SHORT-TERM POWER | RISK ANALYTICS

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PROFILE

ML/DL researcher developing forecasts and analytical tools for ENGIE's short-term power trading desk, with experience in front-office market risk, pricing and market microstructure. Seeking Citi's 2027 Markets Sales and Trading off-cycle internship in Paris.

Available from January 2027 for six months | University internship agreement confirmed.

EDUCATION

Université Paris 1 Panthéon-Sorbonne - M.Sc. Data Science 2025-2026

Machine learning, deep learning, statistics, optimisation and time series

Université Paris Cité - M.Sc. Applied Economics - Econometrics 2023-2025

Econometrics, statistical inference, financial modelling and panel data

Université Paris Cité - B.Sc. Economics & Mathematics 2020-2023

PROFESSIONAL EXPERIENCE

ENGIE Global Markets - ML/DL Research Intern - Power Market Jun 2026-Present

Global Market Analysis, Paris | Research and tools for the short-term power trading desk

- **Desk analytics:** Developed NYX for day-ahead power-price forecasting, combining Chronos-2, CatBoost residual models and adaptive Kalman filtering to support the short-term power trading desk.
- **Forecast evaluation:** Reduced pooled MAE by 8.27% versus Chronos-2 alone in a retrospective 350-day evaluation across four European markets; assessed price spikes, negative prices and forecast uncertainty.
- **Power fundamentals:** Analysed residual load, cross-border flows, nuclear availability and supply stacks to explain day-ahead price dynamics under European market coupling.
- **Research communication:** Built rolling validation and calibration workflows; documented NYX methods, results and limitations in an unpublished English research manuscript.

Société Générale - Quant Market Risk Intern Sep 2025-Mar 2026

Front Office Market Risk, Paris

- **Market risk:** Modelled financial time series with GARCH, EGARCH and VAR; supported Greeks, VaR and stress-scenario analysis.
- **Data and controls:** Built Python/SQL/API pipelines for large risk datasets; applied clustering and KNN to anomaly detection and risk monitoring.

Autorité des marchés financiers (AMF) - Data Scientist / Economist Jan-Jun 2025

Financial Stability & Risk Studies, Paris

- **Market microstructure:** Analysed millions of MiFID II transactions using PySpark, Python, R and panel econometrics to study investor behaviour and market dynamics.
- **Research output:** Engineered investor-level turnover, return and trading-intensity indicators; contributed empirical analysis to the AMF Market and Risk Mapping 2025.

Generali - Quantitative Analyst Intern Apr-Sep 2024

ALM & Strategic Asset Allocation, Paris

- **Pricing:** Built XGBoost surrogate models for computationally intensive full-revaluation pricing; investigated non-linear relationships between risk factors.
- **Rates and analytics:** Calibrated Vasicek/CIR rate models and automated risk and ALM workflows with Python, SQL and Power BI.

MARKETS AND TECHNICAL SKILLS

Markets and risk: Power, oil, derivatives, rates, pricing, Greeks, VaR/stress testing, ALM, market microstructure

Programming and data: Python, SQL, R, PySpark/Spark, Git, pandas, NumPy, statsmodels, Power BI

ML/DL: PyTorch, TensorFlow, scikit-learn, CatBoost, XGBoost

Methods: Time-series forecasting, econometrics, Monte Carlo, optimisation, backtesting, anomaly detection

Languages: French (native), English (C1), Spanish (beginner)